

PROJECT TRICORDER

Mk I · Field Operations Manual · Rev 0.6.28

On-device multimodal field instrument
Inquiry · vision · imagine · speech in/out · translator · audio watch
flora/fauna · sentry · dictation · readings · overhead cards · Memory Alpha · MEDICAL

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RENT-A-HAL Android Edition · software 0.6.28 · 13 September 2026
Supersedes Rev 0.6.24 (12 Sep 2026)*

1. Purpose and scope

This is the operator and installer document for Project Tricorder Mk I through software 0.6.28. It supersedes Rev 0.6.24. Material from 0.6.4–0.6.24 remains in force unless a later row in §2 replaces it. The on-device Help sheet is a short briefing. The HTML briefing shipped in the APK (manual.html) matches this revision. This Word file is the full operations manual.

NAVIGATION WARNING. The artificial horizon, G-meter, GPS speed, barometric altitude proxy, satellite plot, star map, SOL orbits, and HF/VHF/UHF forecasts are NOT certified by the FAA, NASA, EASA, ICAO, or any civil or military authority. They MUST NEVER be used as primary or backup navigation, attitude, or approach references for any aircraft, rotorcraft, UAV, spacecraft, or surface vessel.

SENTRY / AUDIO WATCH WARNING. Weapon, gunfire, and conflict alerts are keyword hits on SmolVLM captions or YAMNet AudioSet labels. They are not a weapons detector, not a 911 substitute, and will false-positive on toys, pictures, fireworks, and media.

KNOWLEDGE / MEDICAL NOTE. MEMORY ALPHA and MEDICAL are field briefings drawn from Wikipedia / WikiMed catalogs over Earth link and cached locally. They are not a physician, not a pharmacopeia, and not Memory Alpha the Star Trek wiki. Treat MEDICAL extracts as orientation only.

DICTATION NOTE. Dictation uses the same on-device Android SpeechRecognizer as Inquiry. Titles, body text, and command words are only as good as that recognizer. Speak command verbs in isolation when possible.

2. What changed since Rev 0.6.24

Rev 0.6.24 closed readings-chart polish and Help/Settings wiring. Builds 0.6.25–0.6.28 add a pinned overhead flight card with photos and city/airframe dossiers, plus voice-only knowledge banks.

Build	Change
0.6.25	Readings bar tap now opens a filtered field-log timeline on the same sheet (category-only playback), not just a spoken summary.

0.6.26	Overhead flight card: OpenSky + adsbdb type/owner/tail. Photo from Planespotters / airport-data / JetPhotos. Snap stored in the field log. Touch photo for airframe facts. Touch FROM/TO for city dossier (Wikipedia + REST Countries + CIA Factbook). Voice banks MEMORY ALPHA (Simple English Wikipedia) and MEDICAL (WikiMed title catalog).
0.6.27	One-shot voice search: “memory alpha Airbus A320” and “medical puncture wound” search immediately. Bare wake still opens the bank, speaks the prompt, and arms the mic. Flight CRT no longer overwritten by the generic context-alert string. AIRFRAME FACTS button even when the photo is late.
0.6.28	MEDICAL extracts and photographs load from English Wikipedia. mdwiki.org REST is Cloudflare-blocked from the device and is used only as a title catalog. Overhead card is pinned until DISMISS AIRFRAME — audio sense no longer flashes it away. City dossier BACK TO AIRFRAME restores the same craft photo so the other city can be opened. CLOSE on city/airframe sheets also restores a pinned card.

3. Face and spoken overlays

The case stays bounded. Settings and Backend are not strips under the instrument. They open as overlay sheets from spoken SETTINGS or BACK END / BACKEND / CAMERA, and from the Help links. XLATE is the translator lamp. ROBO green means robot voice for all Inquiry speech; the letters themselves turn green — there is no extra ON word.

Control	Action
I Inquiry	Chat (Gemma). Wake words: DICTATION, SETTINGS, BACK END, DIAGNOSTICS, HELP, COMPUTER, MEMORY ALPHA, MEDICAL, SUMMARIZE READINGS.
II Sensor	Single-look vision + flora/fauna on the still.
III Sentry	Continuous vision default. Weapon/violence watch. HOLD freezes a frame.
IV Audio	YAMNet + BirdNET + context watches (NOAA, aircraft, civic).
T / XLATE	Offline ML Kit translator. SOURCE / TARGET wheels. AUTO swaps after each line.
H Horizon	Uncertified attitude toy. Toggle on the face.
S Sat/Stars	First press satellites, second stars.
O Sol	1 space weather + orbits + HF/VHF/UHF. 2 local earth forecast. 3 off.

LEVP	Log review, log entry, video log, play.
IPL / ALL	Abort in-flight chat, vision, imagine, voice, translate.
STT slider	Kill only Kokoro / in-flight speech.
ROBO	Green = cheap robot voice for Inquiry speech.

4. Knowledge banks — MEMORY ALPHA and MEDICAL

4.1 Intent

These banks exist so the operator can ask the instrument a reference question without leaving Inquiry and without adding a fifth Termux port. They are voice-only. They are not buttons on the face. That is deliberate: the illusion is an omnipotent computer, not another menu.

4.2 How to use

Arm Inquiry. Press the mic. Either:

One-shot (preferred). Speak the bank and the topic in one utterance.

memory alpha Airbus A320

medical pink eye

medical puncture wound

Two-step. Say only MEMORY ALPHA or MEDICAL. The bank opens, the robot says it is online, and the mic arms about 1.6 s later. Then speak the topic. Say TERMINATE or tap CLOSE to leave. A SEARCH field on the sheet accepts typed queries.

4.3 What is actually fetched

Bank	Titles	Extract + photograph	Cache
MEMORY ALPHA	Simple English Wikipedia opensearch; English Wikipedia if empty	Simple English REST summary, then English Wikipedia query extracts + pageimage	files/knowledge/articles
MEDICAL	mdwiki.org opensearch (WikiMed catalog) plus English Wikipedia opensearch	English Wikipedia REST / query extracts. mdwiki REST summaries are Cloudflare-blocked from the APK and are not used for body text.	same

There is no ~30 MB Simple English Wikipedia ZIM with images. Current Kiwix editions are large (simple-all-mini ≈ 434 MB with images; WikiMed mini ≈ 155 MB) and are not auto-downloaded at init. The voice banks work as soon as Earth link (8.8.8.8) is up and cache each article locally.

4.4 What MEDICAL is not

MEDICAL is a field orientation layer. It will describe pink eye, puncture wounds, and fever in encyclopedia language. It will not diagnose, dose, or triage. If the operator needs care, they need a clinician, not a tricorder extract.

5. Overhead flight card

5.1 Arming

IV Audio Sense with the aircraft checkbox on (default ON) and Earth link up. OpenSky is queried in a box around the last GPS fix. Grounded tracks are ignored. At most two new airborne contacts are announced per poll.

5.2 What the card shows

Callsign, registration, type, owner, spoken altitude/speed (metric or English per Settings), and FROM / TO when adsbdb or OpenSky published a route. A photograph is requested in this order: adsbdb thumbnail, Planespotters hex/reg, airport-data.com Mode-S thumb, unofficial JetPhotos worker. The JPEG is stored on the field-log snap.

5.3 Touch targets

Control	Result
FROM / TO city	City dossier sheet: Wikipedia extract, flag/capital/population when REST Countries resolves a country, CIA World Factbook snippet. Spoken.
Photo or AIRFRAME FACTS	Airframe sheet: registration, ICAO type, manufacturer, owner, country, Mode S, Wikipedia type extract and thumb.
BACK TO AIRFRAME	On the city sheet. Closes the dossier and restores the same craft card and photo so the other city can be opened.
DISMISS AIRFRAME	On the CRT card. Releases the pin. Audio sense may write the CRT again.
CLOSE on city/airframe sheets	If the card is still pinned, the CRT card is restored.

5.4 Pin behavior (0.6.28)

Before 0.6.28 the YAMNet cycle called setScreen every dwell and wiped the photograph. The card is now pinned. Audio sense still classifies, still logs, and may still speak gunfire/siren/vehicle lines. It does not steal the CRT until DISMISS AIRFRAME. The generic “local tricorder detected flight event...” string is not painted over the card.

5.5 Speech format

Robot voice, not Kokoro. Settings toggle metric vs English. Example English: tail ID N2AAA, 11 338 feet, 358 miles per hour, from Newark EWR to Boston BOS. Periods that would stall Kokoro are stripped when Kokoro is used elsewhere; flight watch does not use Kokoro.

6. Readings chart (retained + 0.6.25)

Say SUMMARIZE READINGS [category] [day|week|month|all]. Omitting category or timeframe defaults to all categories for today. Letter bars A B C with a QTY label, color legend, and totals in the lower pane. Touch a bar to speak that category and open the filtered log timeline on the same sheet (0.6.25). Day / Week / Month / All chips are live.

7. Dictation (retained from 0.6.15–0.6.21)

Wake: dictation / dictation mode / start dictation. Directory verbs: NEW, EDIT nnnn, LOG nnnn (CONFIRMED appends the memo to the field log), SUMMARIZE nnnn, MINDMAP keyword, SPEAK nnnn, KILL nnnn (CONFIRMED), TERMINATE. ZULU then SAVE or ABORT (ABORT needs CONFIRMED).

Token buffer: CURSOR MINUS word, CURSOR PLUS word, BACKSPACE, NEWLINE, CURSOR CUT / CURSOR CUT WORD, CURSOR PASTE, CURSOR HOME / END. Punctuation: PUNCTUATION PERIOD / COMMA / SEMICOLON / QUESTION; bare PERIOD and DOT also attach a mark to the word before the cursor.

Mindmap uses synset plus 128-d feature hashes. “puzzle” also hits riddle and crossword. The Universal Sentence Encoder file may fetch on first Earth-link boot; matching still works offline without it.

8. Sentry and audio watch (retained)

III captions every dwell. Motion does not gate the look. Sure-weapon tokens (gun, pistol, rifle, knife, and listed arms) raise klaxon + WEAPON DETECTED ×3 on the robot voice. Dual-use / possible violence raises a chime. Continuous vision (no Kokoro) is the default sentry path so the watch is not waiting on neural TTS.

IV maps YAMNet labels through a priority table: gunfire ×3 + klaxon; screaming/shouting; siren ×3; air / rail / ship; vehicles; named animal; humans ×3; entertainment device. Capture is muted while the unit itself is talking so flight announcements are not logged as humans detected. Fireworks can look like gunfire.

9. IPL, Imagine, translator, horizon, sky

IPL / ALL aborts in-flight chat, vision, Imagine, voice, and translate. The STT slider kills only speech. Text longer than 80 characters is chunked to Kokoro. A Settings slider 400–8000 caps robot-TTS length; the toggle disables the cap.

Imagine: step and size sliders (256 / 384 / 512) against SD Turbo on :8188. Offline pictures are a field pictogram, not a toy — the documented jungle use case is “imagine a cave entrance” while the translator asks the local about caves.

Translator: ML Kit packs, SOURCE / TARGET wheels, AUTO swap after each line. XLATE lamp green idle, yellow speaking, red thinking. Horizon / Sat / Stars / Sol as on the face. Uncertified. Sol third press turns the panel off. Earth forecast scrolls.

10. Field log kinds

Kind	When written
sensor / sentry	Still + caption. Flora line when the pack is on device.
audio_*	YAMNet / BirdNET watch hits.
flight	Overhead contact + photo JPEG + route extra.
flora	iNaturalist pack hits on a Sensor still.
officer	Spoken LOG ENTRY or LOG nnnn from dictation.
video	Front-camera clip when VIDEO LOG is armed.
context	NOAA / civic / remote mesh events.

11. Architecture

Kotlin WebView APK. Four Termux loopback services stay up together: Gemma chat :8080, SmolVLM vision :8081, SD Turbo imagine :8188, Kokoro TTS :5000. Launch them with `~/bin/tricorder start (sshd on 8022)`. Do not stop 8188 to free a port.

On-device, no extra port: YAMNet, BirdNET, iNaturalist flora/fauna (TFLite), ML Kit translator, Android SpeechRecognizer, Android TTS robot voice, Wikipedia/WikiMed REST + local article cache, OpenSky / adsbdb / Planespotters, NWS CAP, Nearby Connections mesh.

12. Install and first run

Uninstall the previous Mk I package before sideloading 0.6.28 (same applicationId). Allow Camera, Microphone, Location, Nearby devices. Termux from F-Droid, not Play. On first Earth-link boot the APK fetches BirdNET, flora packs, a translator pair, TLEs, SOL weather, and warms the knowledge banks. It cannot fetch the four GGUF/ONNX weights; those remain the Termux script already documented.

13. Troubleshooting additions for 0.6.25–0.6.28

Symptom	Likely cause	Action
MEMORY ALPHA opens, no search	Pre-0.6.27 ate the rest of the sentence and did not arm the mic.	Sideload 0.6.27+. Say the topic in the same utterance.
MEDICAL titles only, no extract or photo	mdwiki REST blocked by Cloudflare.	0.6.28 pulls body and image from English Wikipedia. Needs Earth link.
Flight photo flashes away	Audio sense setScreen every dwell.	0.6.28 pin. Use DISMISS AIRFRAME when done.
No FROM/TO	adsbdb/OpenSky have no route for	Card still shows type/photo when

	that callsign.	those lookups hit.
No overhead at all	Aircraft checkbox off, no GPS, OpenSky empty box, or no Earth link.	IV Audio + aircraft ON + Wi-Fi + outdoors.
City BACK missing	Card was dismissed, or lastFlightCard never set.	Open FROM/TO from a live pinned card.
Knowledge sheet ignores later speech	TERMINATE / CLOSE cleared the bank, or Inquiry is not armed.	Say MEMORY ALPHA again or use the SEARCH field.
Lights red, backends up	WebView cannot reach 127.0.0.1.	Confirm ~/bin/tricorder status. Do not kill ports.

14. Competing products (unchanged conclusion)

Phone “AI” products in 2026 are almost all cloud chat with optional on-device speech. Note-takers run Whisper then a summarizer. Pocket-sized LLM apps run one GGUF. None combine Gemma + SmolVLM + SD Turbo + Kokoro + YAMNet/BirdNET/flora + offline translation + sentry/audio watch + mesh + overhead photography + spoken knowledge banks on one offline Snapdragon face. That combination is the point of Mk I.

15. Revision history

Rev	Date	Note
0.6.14	9 Sep 2026	Audio soundscape watch.
0.6.17	9 Sep 2026	Dictation token buffer.
0.6.21	10 Sep 2026	Flight route speech, ROBO, DIAGNOSTICS.
0.6.24	12 Sep 2026	Readings letters, TTS mute, overlay Settings.
0.6.26	13 Sep 2026	Flight photos, city/airframe facts, knowledge banks.
0.6.27	13 Sep 2026	One-shot knowledge search; flight card not clobbered.
0.6.28	13 Sep 2026	MEDICAL extracts; pinned airframe; city BACK.

End of Rev 0.6.28. Keep this file with the APK and the project source zip. Project Tricorder is not authorized by or affiliated with The Wand Company, Paramount SkyDance, Desilu, Gulf and Western, CBS, Roddenberry Entertainment, or Wah Chang, which are trademarks of their respective holders.